```
                dlon = (dist_m / (111320.0 * scale)) * math.sin(heading_rad)

                pred_lat = lat + dlat
                pred_lon = lon + dlon
                pred_alt_m = max(0.0, altitude_m + (vs_cms / 100.0) * age_sec)

            predicted.append(
                {
                    "icaoAddress": icao,
                    "callsign": str(r["callsign"] or "UNKNOWN").strip().upper() or "UNKNOWN",
                    "latDD": pred_lat,
                    "lonDD": pred_lon,
                    "altitudeMM": int(round(pred_alt_m * 1000.0)),
                    "headingDE2": heading_de2,
                    "horVelocityCMS": speed_cms,
                    "verVelocityCMS": vs_cms,
                    "timestamp": now_dt.isoformat(),
                    "sourceIp": "PREDICTED",
                    "isPredicted": True,
                    "predictionAgeSec": int(round(age_sec)),
                }
            )

        return predicted

    def _append_icao_excel_rows(self, rows):
        """
        Create/update one Excel file per ICAO with continuous append.
        File format: <ICAO>.xlsx
        """
        if not rows:
            return

        headers = list(self._master_headers)

        grouped_rows = defaultdict(list)
        for row in rows:
            record = self._to_icao_excel_record(row)
            if record is None:
                continue
            grouped_rows[record["icao"]].append(record)

        for icao, icao_rows in grouped_rows.items():
            file_path = self.icao_excel_dir / f"{icao}.{self.icao_excel_extension}"
            try:
                if file_path.exists():
                    wb = self._load_workbook_path(file_path)
                    ws = wb.active
                else:
                    wb = Workbook()
                    ws = wb.active
                    ws.title = "flight_data"

                self._ensure_excel_headers(ws, headers)
                for record in icao_rows:
                    ws.append(
                        [
                            record["timestamp"],
                            record["icao"],
                            record["callsign"],
                            record["latitude"],
                            record["longitude"],
                            record["altitude"],
                            record["heading"],
                            record["speed"],
                            record["vertical_rate"],
                        ]
                    )
                    ws.cell(row=ws.max_row, column=4).number_format = "0.00000000"
                    ws.cell(row=ws.max_row, column=5).number_format = "0.00000000"
                self._save_workbook_path(wb, file_path)
            except Exception:
                # Keep logging uninterrupted for other ICAO files.
```